3) Use a screwdriver to unhook the locking lever of the right locking plate and pull the locking plate off of the bus module.

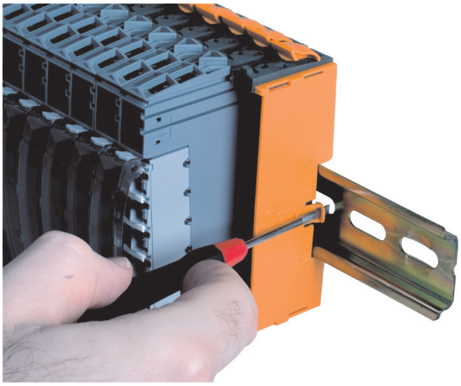


Figure 215: Unlock the right locking plate with a screwdriver

4) Now you can add more modules as described for assembly method 2 (see section 2.2 "Method 2" on page 430).

6. Installation of accessories

6.1 Additional locking mechanisms

Some specific areas require additional locking mechanisms to prevent accidental release of the mechanics.

6.1.1 Additional locking clip

The additional clip attaches the electronic module to the bus module. The locking clip is inserted in the appropriate opening on the module and pushed down.

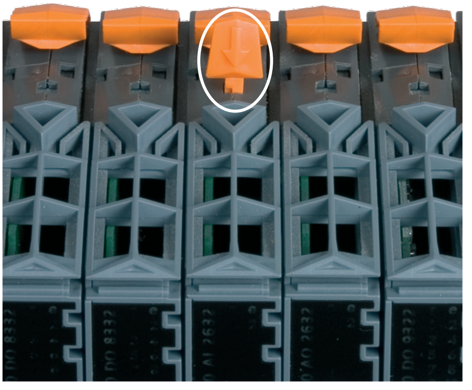


Figure 216: Installing the additional locking clip

6.1.2 Terminal locking clip

The terminal locking clip attaches the terminal block securely to the electronic module.

1) Set the terminal locking clip on the terminal block locking lever as shown.

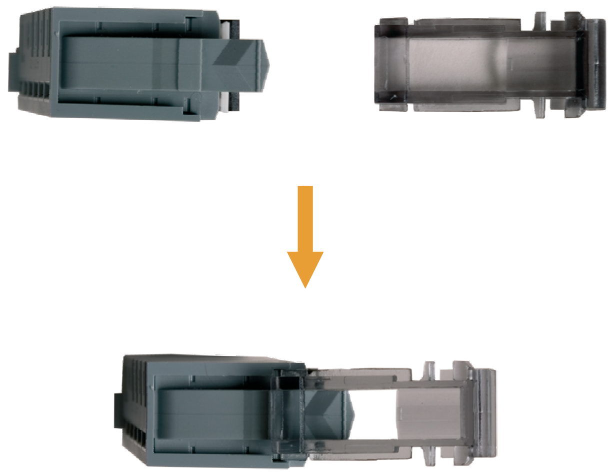


Figure 217: Set the terminal locking clip on the terminal block locking lever

2) Push down and hold the terminal locking clip and the locking lever with your index finger ①. Finally, slide the terminal locking clip forward with your thumb ②.

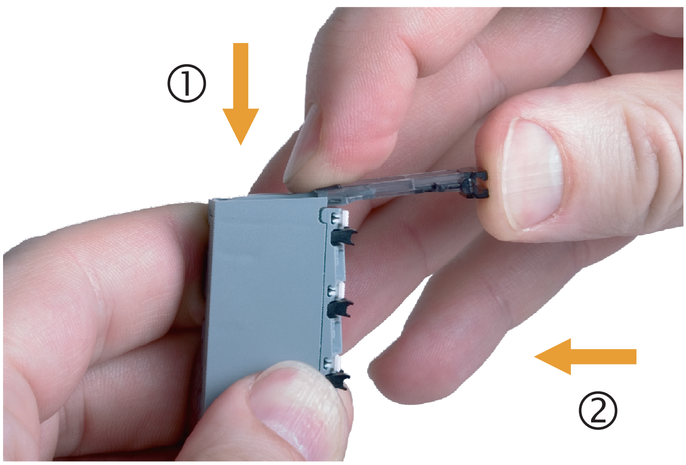


Figure 218: Install terminal locking clip on terminal block

3) Hang the bottom edge of the terminal block in its place on the bus module.

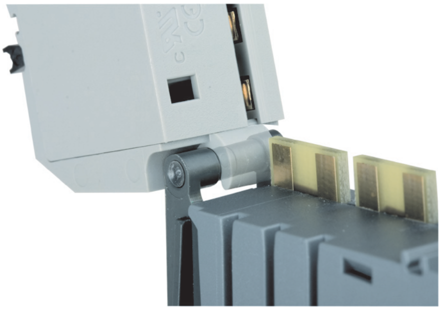


Figure 219: Hang the bottom edge of the terminal block in its place on the bus module

4) Rotate the terminal block up into place.

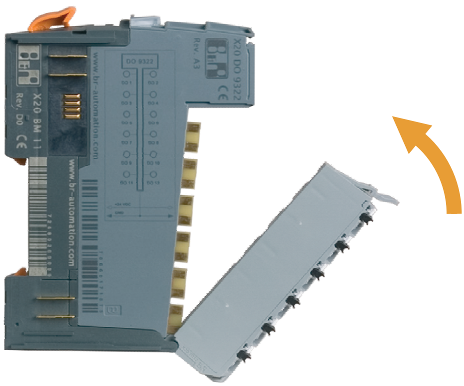


Figure 220: Rotate the terminal block up into place

5) Secure the terminal block in the electronic module by pushing in the terminal locking clip.

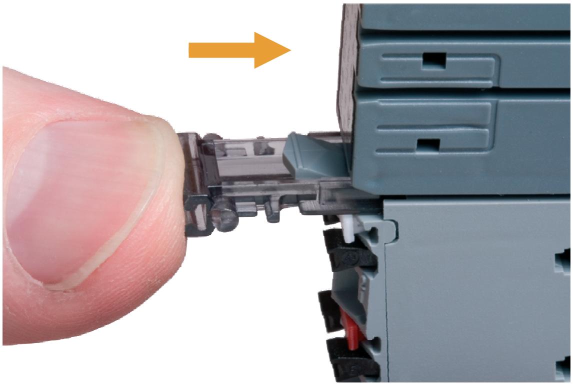


Figure 221: Secure the terminal block in the electronic module

6) Installed terminal locking clip.

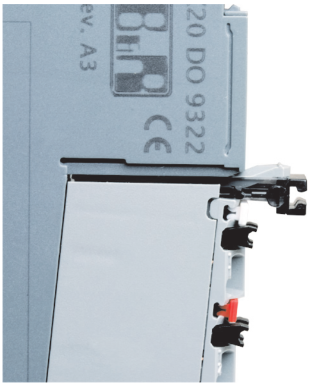


Figure 222: Installed terminal locking clip

7) To remove the terminal block, pull the terminal locking clip out again.

6.2 Plain text cover

Covers are available for X20 modules into which plain text legend strips can be inserted. The covers are attached to the terminal locking clips.

1) Hold the plain text cover at a 90° angle to the terminal locking clip.

2) Push the plain text cover into the terminal locking clip until it clicks into the slot on the clip.

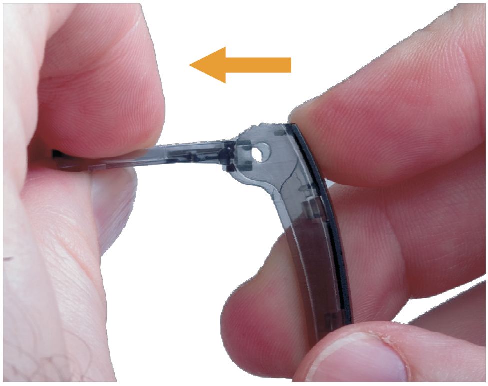
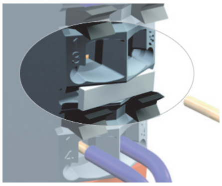
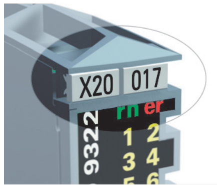
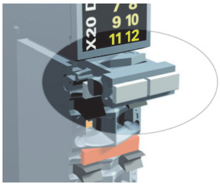
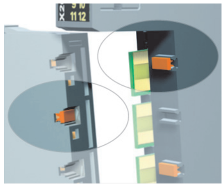


Figure 223: Attach plain text cover to terminal locking clip

7. Label tabs

The label tabs can serve the following purposes:

	Labeling the terminals		Labeling the module
	Labeling the terminal blocks		Coding the terminals

The labeling tool is needed to attach the label tabs.

Figure 224: Labeling tool

7.1 Labeling the terminals

This section explains how to label the terminals. The terminal blocks and the modules are labeled in a similar manner.

Figure 225: Terminal block with label tabs

1) Grip the desired label tabs with the double-width cutters of the labeling tool.

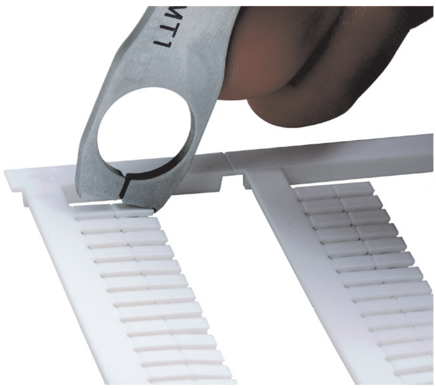


Figure 226: Grip desired label tabs with labeling tool

2) Press with the labeling tool to separate the labels.

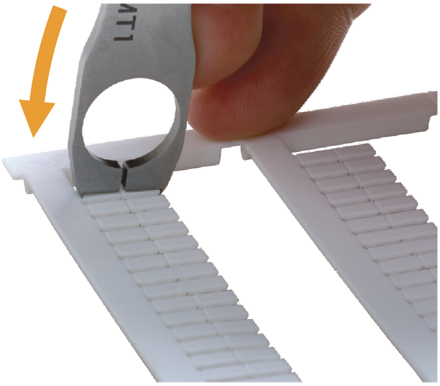


Figure 227: Separate labels with the labeling tool

3) Center the label tabs over the slot on the terminal block.

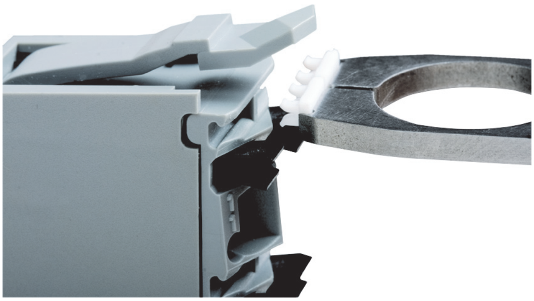


Figure 228: Center label tabs over slot

4) Hold the labeling tool at approximately an 80° angle to the terminal block.

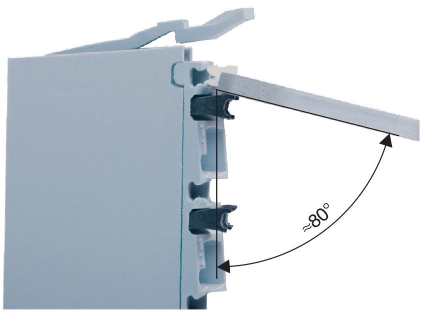


Figure 229: Hold the labeling tool at approximately 80° angle to the terminal block

5) Press with the labeling tool to insert the feet of the label tabs into the slot.

6) Inserted label.

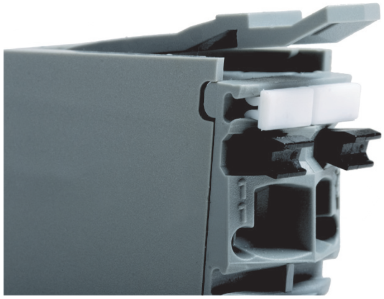


Figure 230: Inserted label

7.2 Coding the terminals

To prevent errors, the X20 terminal blocks can be coded. This helps prevent terminal blocks from being inserted in the wrong electronic module.

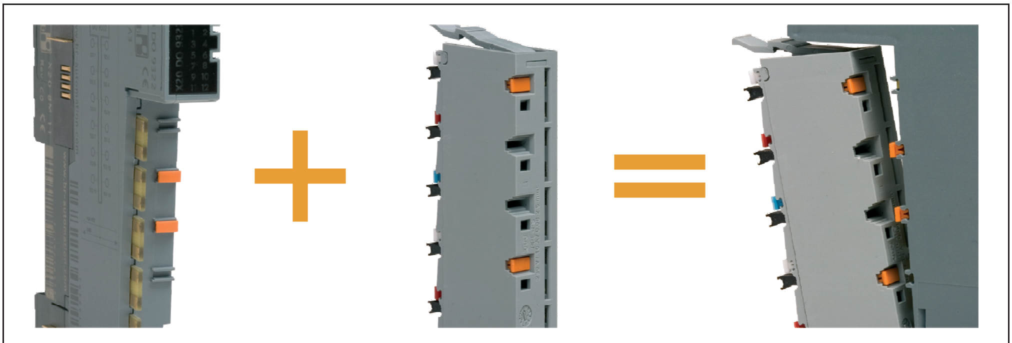


Figure 231: Terminal coding helps prevent errors from the start

1) Grip the desired label tab with the single-width cutters of the labeling tool (compare with section 7.1 "Labeling the terminals" on page 448).

2) Center the label tab over the slot on the electronic module.

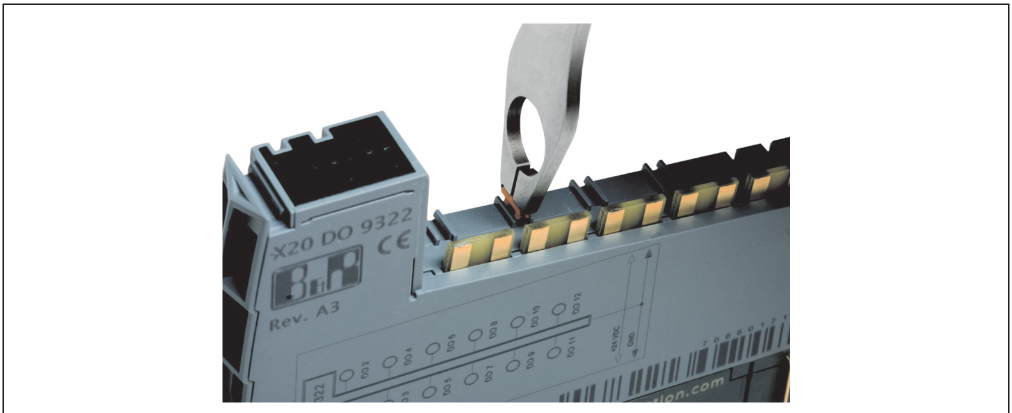


Figure 232: Center the label tab over the slot on the electronic module

3) Hold the labeling tool at a 90° angle to the electronic module and press to insert the label's feet into the slot.

4) Remove a label tab with the single-width cutter of the labeling tool.

5) Set the label tab in the slot on the back of the terminal block as shown.

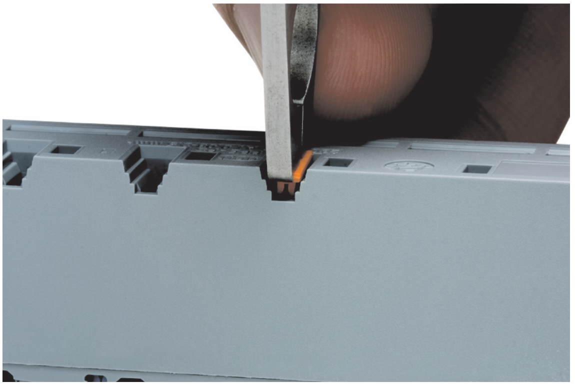


Figure 233: Set the label tab in the slot on the back of the terminal block

6) Use the labeling tool to push the left feet of the label into the slot.

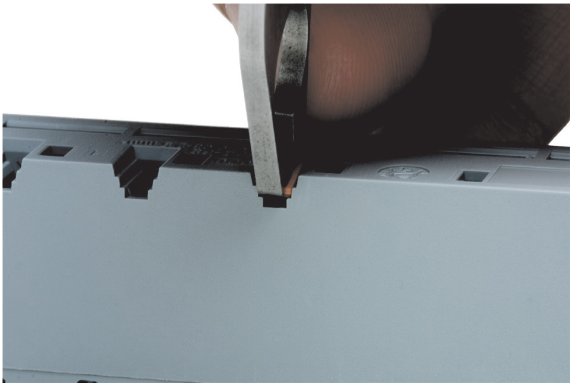


Figure 234: Press left feet of the label into the holes

7) With the labeling tool, press the right feet of the label into the slot.

Figure 235: Press right feet of the label into the slot

8) Inserted label for terminal coding.

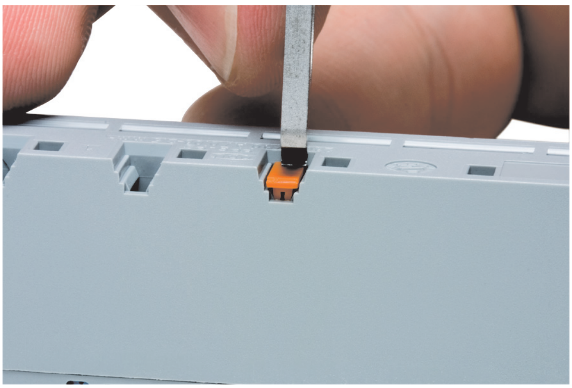


Figure 236: Inserted label for terminal coding

Chapter 7 • Standards and certifications

1. Standards and limits used

The product standard EN 61131-2 is generally valid for B&R industrial products (identical with IEC 61131-2). The following standards provide a detailed definition of proper functionality in a typical industrial environment (charged with electromagnetic energy):

Standard	Description
EN 61000-6-4	Electromagnetic compatibility (EMC); Generic standard - emission standard Part 2: Industrial environment (EN 50081-2 will be replaced by IEC 61000-6-4)
IEC/CISPR 11	Industrial, scientific and medical high-frequency device radio disturbances limits and measuring procedures
EN 61000-6-2	Electromagnetic compatibility (EMC) - Generic standard, emission standard Part 2: Industrial environment (EN 50082-2 will be replaced by IEC 61000-6-2)
EN 61131-2 Edition 2	Programmable logic controllers Part 2: Equipment requirements and tests
UL 508	Industrial control equipment, (UL = Underwriters Laboratories)
EN 60204-1	Safety of machinery - electrical equipment on machines Part 1: General requirements
EN 60529	IP20 protection

Table 230: Overview of standards

1.1 Tests, limits, and test subject behavior

Criteria	Description
A	Uninterrupted operation during the test.
B	Brief interruption during the test allowed.

Table 231: The criteria A and B describe the test subject behavior

EN 61000-4-2, electrostatic discharge	
	IEC 61131-2 Ed. 2
Contact discharge to powder-coated and bare metal parts	±4 kV, criteria B
Air discharge to plastic parts	±8 kV, criteria B

Table 232: Limits for electrostatic discharge

EN 61000-4-3, electromagnetic fields	
Housing, completely wired	80 MHz - 2 GHz, 10 V/m, criteria A 80% amplitude modulation at 1 kHz

Table 233: Limits for electromagnetic fields

EN 61000-4-4, burst, fast transients	
	IEC 61131-2 Edition 2
Power supplies and AC inputs/outputs	2 kV, 1 min, criteria B
Data cable	1 kV, 1 min, criteria B
Digital inputs/outputs	1 kV, 1 min, criteria B
Analog inputs/outputs	1 kV, 1 min, criteria B

Table 234: Limits for fast transients

EN 61000-4-5, surge		
	Common mode, unsymmetrical	Differential mode, symmetrical
AC power supplies	2 kV (12 Ω), criteria B	1 kV (2 Ω), criteria B
DC power supplies	1 kV (12 Ω), criteria B	0.5 kV (2 Ω), criteria B
Digital and analog I/O, AC, unshielded AC auxiliary voltage outputs for sensors, etc.	2 kV (42 Ω), criteria B	1 kV (42 Ω), criteria B
Digital and analog I/O, DC, unshielded Data lines, unshielded DC auxiliary voltage outputs for sensors, etc.	0.5 kV (42 Ω), criteria B	0.5 kV (42 Ω), criteria B
All shielded lines	1 kV (2 Ω), criteria B	---

Table 235: Limits for surge

EN 61000-4-6, conducted disturbances (radio frequency)	
Network I/O Signal connections >10 m Functional ground connection	150 kHz - 80 MHz, 10 V, criteria A 80% amplitude modulation at 1 kHz

Table 236: Limits for conducted disturbances (radio frequency)

EN 55011 (IEC/CISPR 11, modified), emission	
	IEC 61131-2 Edition 2
30 MHz - 230 MHz	40 dB (µV/m), quasi-peak value, measured at distance of 10 m
230 MHz - 1 GHz	47 dB (µV/m), quasi-peak value, measured at distance of 10 m

Table 237: Emission

EN 60664-1, pollution degree
Pollution degree 2: non-conductive material

Table 238: Degree of pollution

EN 60068-2-6, vibration		
Frequency range [Hz]	Continuous	Periodic
$5 \leq f < 9$	1.75 mm amplitude	3.5 mm amplitude
$9 \leq f \leq 150$	0.5 g constant acceleration	1 g constant acceleration
$f > 150$	Not defined	Not defined

Table 239: Limit values for vibration

EN 60068-2-27, shock
15 g over 11 ms, half sinus wave, 3 shocks per axis and direction (total 18 shocks)

Table 240: Limit values for shock

2. International standards

B&R products and services comply with all applicable standards. They are international standards from organizations such as ISO, IEC and CENELEC, as well as national standards from organizations such as UL, CSA, FCC, VDE, ÖVE, etc. We give special consideration to the reliability of our products in an industrial environment.

Certifications	
USA and Canada	All important B&R products are tested and listed by Underwriters Laboratories and are checked quarterly by a UL inspector. This certification is valid in the USA and Canada and makes it considerably easier to license your machines and systems in these markets.
Europe	All harmonized EN standards for the valid guidelines are met.
Russian Federation	GOST-R certification is available for the export of all B&R products in the Russian Federation. The X20 System is in preparation to be sent in for certification.

Appendix A • Abbreviations

1. General information

Abbreviations appear throughout the User's Manual, for example in data tables or descriptions of pin assignments.

2. Overview

Abbreviation	Stands for	Description
NC	Normally closed	A relay contact. Also known as an "opener".
	Not connected	Used in the description of pin assignments if a terminal or pin is not connected to a module.
ND	Not defined	In data tables, this stands for a value that has not been defined. Because a cable manufacturer does not provide certain technical data, for example.
NO	Normally open	A relay contact. Also known as an "closer".
TBD	To be defined	Used in the technical data tables when certain pieces of information are not yet available. The value will be provided later.

Table 241: Abbreviations used in the User's Manual

Numbers

A

B

C

D

P

R

S

T

V

X

Z

X